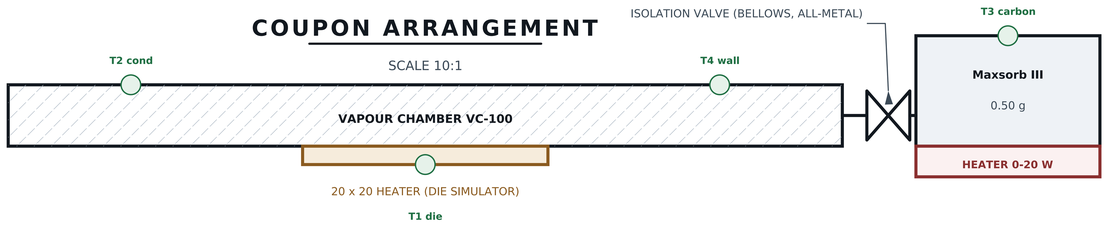
EXPERIMENTAL COUPON — NOT A PRODUCTION CONFIGURATION — SEE PHYSICS NOTE BELOW

COUPON ARRANGEMENT

SCALE 10:1



Mesh: 10 um SUS 316L — NOT 50 um. Maxsorb III fines pass a 50 um mesh and foul the wick.

Carbon bake-out 250 C / 8 h / < 1e-3 Pa before installation, transferred under vacuum.

Reservoir 0.5 g Maxsorb III needs >= 1.8 cm3 at 0.28 g/cc bulk density — the original 1.50 cm3

housing could not physically hold the charge. This coupon uses 30 x 12 x 6 = 2.16 cm3.

ADSORPTION EQUILIBRIUM — DUBININ-ASTAKHOV, METHANOL / MAXSORB III

T evap	T res	P/Psat	UPTAKE	HELD	WICK FILL	RESULT
60 C	40 C	1.00	1.31 g/g	0.66 g	15 %	**DRY**
60 C	60 C	1.00	1.31 g/g	0.66 g	15 %	**DRY**
60 C	100 C	0.24	1.06 g/g	0.53 g	35 %	**DRY**
60 C	127 C	0.11	0.77 g/g	0.39 g	58 %	marginal
60 C	169 C	0.03	0.42 g/g	0.21 g	90 %	wet

PHYSICS NOTE — WHY THE PASSIVE VERSION CANNOT WORK

- In a sealed pure-substance chamber the pressure is set by the evaporator: P = Psat(T_evap). The carbon sees the relative pressure P / Psat(T_res).
- A reservoir on the condenser side always satisfies T_res <= T_evap, so P/Psat >= 1 and the carbon saturates: it takes 0.66 g of a 0.75 g charge and the wick runs at 15 % fill. That is permanent dry-out, not pressure control.
- Keeping the wick wet needs the reservoir ~110 K ABOVE the chip. A working vapour chamber is isothermal, so no passive layout can deliver that. It requires the heater and valve drawn above.
- Overcharging to 1.90 mL keeps the wick wet, but then liquid always remains, the pressure stays at Psat, and the carbon buys nothing thermally. Its only genuine residual function is as a non-condensable gas getter — for which tens of mg suffices, not 0.5 g.

COUPON vs PRODUCTION VC-100

	WITH CARBON	WITHOUT
Wick fill at equilibrium	15 %	**125 %**
Q at 2 m/s forced air	0 W (dry)	**113 W**
Transient buffer	722 J (heater needed)	0 J
Parts count / failure modes	+ valve, heater, control	**baseline**

INSTRUMENTATION

MEASUREMENT	INSTRUMENT	PURPOSE
T1..T4	type-T TC, ± 0.2 K	gradient map
Chamber pressure	capacitance manometer	verify Psat(T)
Reservoir mass	in-situ load cell ± 1 mg	uptake vs time
Heater power	4-wire, ± 0.5 %	energy balance
Valve state	open / closed	isolate the carbon

TEST MATRIX

#	TEST	MEASURES
1	Valve CLOSED, sweep 0-150 W	baseline R th, no carbon
2	Valve OPEN, reservoir unheated	dry-out rate + time constant
3	Valve OPEN, reservoir held 110 K above die	whether the wick recovers
4	Valve OPEN, cyclic 10 W bursts, heater regen	usable buffer energy (722 J predicted)
5	1000 h hold, valve OPEN, unheated	NCG getter benefit vs dry-out cost

COUPON NOTES

- Fluid is METHANOL for this coupon (activated carbon is hydrophobic; water uptake is poor). Accept the 7.3x lower merit number — this coupon measures adsorption, not peak cooling.
- Methanol charge 1.90 mL so the wick stays wet with the carbon saturated. Do not run at 1.00 mL.
- The valve is what makes this an experiment rather than a defect. A production unit has no valve, so it has no way to stop the carbon draining the wick.

IF TEST 4 SUCCEEDS — PATH TO A PRODUCT

- Move the reservoir onto the die's own heat path, upstream of the chamber, so it is genuinely hotter than the evaporator without a valve.
- Replace the bellows valve with a thermally actuated one-way element, or accept a control loop driving the regeneration heater.
- Budget 722 J against ~5 g of added mass and a new hermetic joint. Compare against simply adding 15 K of copper (+50 g).

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE
MAXSORB III RESERVOIR — TEST COUPON
EXPERIMENTAL / NOT FOR PRODUCTION / INSTRUMENTED

SCALE **10:1** — THIRD ANGLE PROJECTION
UNITS **mm**

DRAWING No. **VC-400**
REV **A**
SHEET **7 / 7**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-09